User Guide

# PF-Fatigue v1.0: An Abaqus/CAE Plug-in for Gradient-Consistent Phase-Field Fatigue Analysis

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**Quick start**

1. Build the part and mesh it with CPE4T, CPS4T, CPE3T or C3D8T elements.
2. Create a Coupled temp-displacement step (`Step-1`), the supports, and one load or boundary condition that drives the specimen (for example u2 at the top edge in `Step-1`).
3. Optional: an initial crack, cut into the geometry or defined as an assembly Set (Section 4).
4. Open Plug-ins → PF-Fatigue. The dialog proposes the driving load, its step and, if the model has no section, an automatic section. Enter the material and fatigue constants and the loading.
5. Click Check Model and read the `[OK]`, `[WARN]` and `[ERROR]` lines in the message area.
6. Click Build / Update, submit `PF-Fatigue-Job` from the Job Manager, and plot NT11 (the phase field $\varphi$).

## 1. Function of the plug-in

PF-Fatigue turns an Abaqus/CAE model into a phase-field fatigue model. The user builds the geometry, the mesh, the supports and one driving load or boundary condition. The plug-in then

- creates the PF-Fatigue material (user material constants and state variables) and assigns it to the sections of the base material, or creates a section for all parts if there is none;
- creates or updates a steady-state coupled temperature–displacement step with fixed increments;
- creates the load amplitude and applies it to the driving load or boundary condition;
- requests field output of displacement, reaction force, stress, phase field and state variables;
- holds the phase field at 1 on an initial-crack Set if requested (a crack cut into the geometry needs no action);
- creates the job with the user subroutine, and submits it if requested.

The plug-in offers the two fatigue laws of the article (Table 1).

**Table 1**: Fatigue laws of the plug-in

| Strategy | Use | Subroutine (bundled) |
|---|---|---|
| Cycle-by-cycle | every load cycle resolved; short tests and benchmark problems | `PF_Fatigue_Exact_EnergySplit.f` |
| Envelope (dN) | long lives; each increment | `PF_Fatigue_GC_Envelope_Final.f` |

| Strategy | Use | Subroutine (bundled) |
|---|---|---|
| | represents ΔN cycles at the peak load | |

The temperature degree of freedom is the phase field $\varphi$: 0 intact, 1 fully broken.

## 2. Requirements

- Abaqus/CAE
- A Fortran compiler linked to Abaqus. `abaqus verify -user_std` checks the set-up.
- Coupled temperature–displacement elements CPE4T, CPS4T, CPE3T or C3D8T.
- One consistent unit system, for example N, mm and MPa; $G_c$ is then in N/mm and $l_0$ in mm.

## 3. Installation

1. Close Abaqus/CAE.
2. Remove any older version of PF-Fatigue from the plug-in directory. Renaming it inside the directory is not enough, because Abaqus/CAE also searches the subfolders for `*_plugin.py` files. The plug-in directory is `abaqus_plugins` in the home directory, or the directory set by `plugin_central_dir` in `custom_v6.env`.
3. Copy the folder `PF_Fatigue` into the plug-in directory. It contains `PF_Fatigue_plugin.py`, `PF_FatigueDB.py`, `PF_Fatigue_kernel.py`, `PF_Fatigue_utils.py`, `icons\` and `subroutines\` with the two Fortran files.
4. Start a new Abaqus/CAE session (an open session keeps the old plug-in) and open Plug-ins → PF-Fatigue. The menu entry has a small red icon with a white crack (Fig. 1).

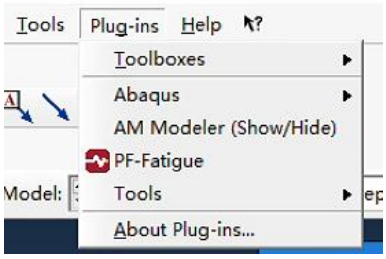



**Fig. 1.** Plug-ins menu with the PF-Fatigue entry.

## 4. Model preparation

Before the plug-in is opened, the model must contain:

1. Part and mesh with one of the supported element types.
2. Material and section (optional). If a homogeneous solid section with a material (for example `Material-1`) is already assigned, the plug-in redirects that section to the PF-Fatigue material and leaves the original material unchanged. If the model has no section, keep Base material = `(auto) section for all parts` (the default in this case). The plug-in then creates the section `PF-Fatigue-Section` (thickness 1 in 2D) and assigns it to every part.
3. Supports: boundary conditions that remove rigid-body motion, usually in the Initial step.
4. Driving load or BC: exactly one load or boundary condition with one non-zero component (a displacement or rotation component, a concentrated force or moment component, or a pressure-type magnitude). It is created in an analysis step, for

example the coupled temperature–displacement step `Step-1`. Its value is replaced by the maximum value entered in the plug-in.

5. Initial crack (optional). Two ways are possible, and both were tested:
    - Geometric crack: a seam (the edge is assigned as a seam before meshing) or a notch cut out of the geometry. Leave Initial crack unticked. Refine the mesh at the tip.
    - Phase-field crack: an assembly Set (edges, faces or nodes) on which the plug-in holds the phase field at 1. Tick Initial crack in the dialog and select the Set.

## 5. Dialog

The dialog has four tabs: Model, Fatigue Loading, Check and Run. The buttons at the bottom are Build / Update, Close and Check Model. The dialog stays open after Build / Update, so settings can be changed and built again. Inputs that the selected strategy does not use are greyed out, for example the envelope constants when Strategy = Cycle-by-cycle.

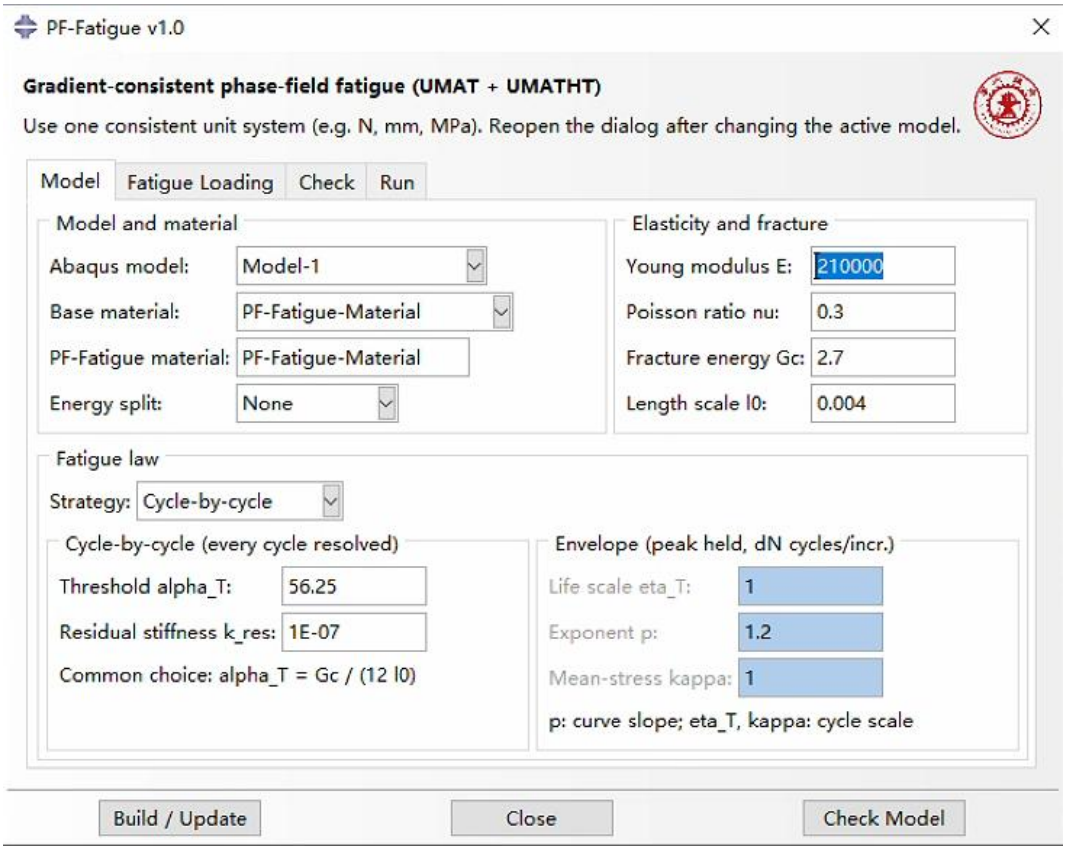



**Fig. 2.** Model tab with Strategy = Cycle-by-cycle; the envelope constants are greyed out. The base material is the PF-Fatigue material because the model had been built once before.

### 5.1 Model tab

The default values are those of the SENT example of the article (steel-like card, $l0$ = 0.004 mm). They must be replaced by the values of the user's material and mesh. Table 2 lists the fields.

**Table 2**: Fields of the Model tab

| Group | Field | Meaning | Typical value |
|---|---|---|---|
| Model and material | Abaqus model | model to modify | current model |
| | Base material | material whose sections are redirected, or `(auto) section` | chosen automatically |

| Group | Field | Meaning | Typical value |
|---|---|---|---|
| | | `for all parts` when the model has no section | |
| | PF-Fatigue material | name of the material created by the plug-in | `PF-Fatigue-Material` |
| | Energy split | split of the crack driving energy: None, Amor, Miehe, No-tension | Miehe for mixed mode |
| Elasticity and fracture | Young modulus E, Poisson ratio nu | elastic constants | 210000, 0.3 |
| | Fracture energy Gc | critical energy release rate | 2.7 (N/mm) |
| | Length scale l0 | phase-field length scale l0 | h $\leq l_0/2$ on the crack path |
| Fatigue law | Strategy | Cycle-by-cycle or Envelope (dN) | Section 6 |
| Cycle-by-cycle | Threshold alpha_T | fatigue threshold of the accumulated energy | Gc/(12$l_0$) |
| | Residual stiffness k_res | residual stiffness | 1e-7 |
| Envelope | Life scale eta_T | dimensionless fatigue threshold; sets the level of the S–N curve and of the crack-growth curve | from identification |
| | Exponent p | sets the slope of the S–N and Paris curves | from identification |
| | Mean-stress kappa | sets the spacing between the curves of different stress ratios (Walker-type parameter) | from identification |

### 5.2 Fatigue Loading tab

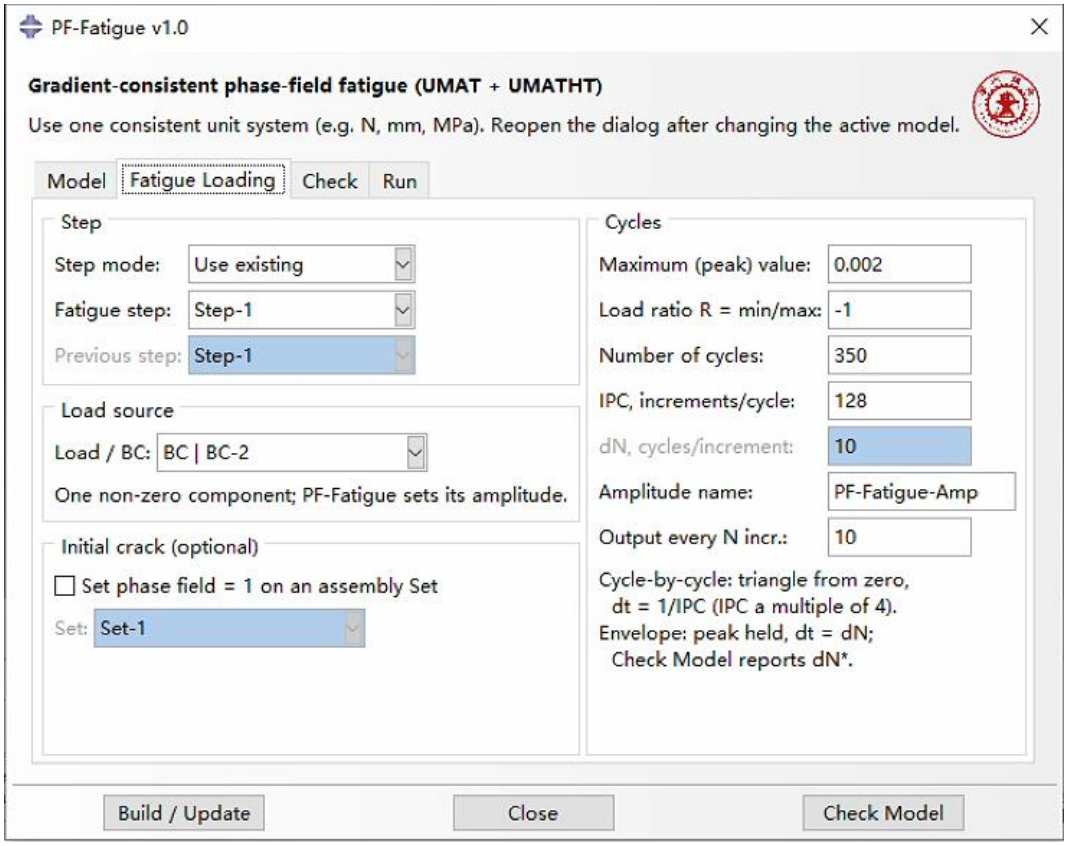



**Fig. 3.** Fatigue Loading tab. dN is greyed out because the strategy is Cycle-by-cycle; the Set list is active only when the initial-crack box is ticked.

When the dialog opens, it selects the first Load/BC with exactly one non-zero component (supports, whose components are all zero, are skipped). It sets Step mode = Use existing and selects the step in which this Load/BC acts. These choices should be checked before continuing. Table 3 lists the fields.

**Table 3**: Fields of the Fatigue Loading tab

| Group | Field | Meaning |
|---|---|---|
| Step | Step mode | Use existing: configure the step that holds the driving load (recommended). Create new: create a new step after Previous step (Previous step is active only in this mode). |
| | Fatigue step / Previous step | step names |
| Load source | Load / BC | the driving object, shown as `BC \| name` or `LOAD \| name`; the list is empty until the model has a load or BC |
| Initial crack (optional) | Set phase field = 1 … / Set | tick, then select the assembly Set on which $\varphi$ is held at 1 |
| Cycles | Maximum (peak) value | value of the driving component at the load peak |
| | Load ratio R = min/max | R < 1 |
| | Number of cycles | cycles to simulate (cycle-by-cycle: whole cycles, a fractional part is dropped; envelope: an integer multiple of dN) |
| | IPC, increments/cycle (cycle- | a multiple of 4 |

| Group | Field | Meaning |
|---|---|---|
| | by-cycle) | |
| | dN, cycles/increment (envelope) | cycles represented by one increment |
| | Amplitude name | name of the created amplitude |
| | Output every N incr. | field-output frequency in increments |

The plug-in generates

- for Cycle-by-cycle: a triangular amplitude that starts from zero, with a quarter-cycle ramp to the maximum and then cycles between max and R·max; fixed increment 1/IPC; step time = cycles + 0.25. The SENT-II input file of the package uses a different start (a 0–1–0 wave that reaches the first peak after half a cycle), so the two are not identical increment by increment;
- for Envelope: a constant amplitude equal to 1 (the peak load is held); fixed increment ΔN; step time = number of cycles. R enters only the material constants.

### 5.3 Check tab and Check Model

Check Model runs all checks and does not change the model. The messages appear in the message area (Table 4).

**Table 4**: Messages of Check Model

| Message | Meaning and action |
|---|---|
| `[OK] Bundled subroutine ... verified` | the Fortran file is the released one |
| `[OK] Scaled cycle increment dN* = ...` | resolution of the envelope law (Section 6.2) |
| `[WARN] dN* = ... exceeds 1.0` | reduce dN; the crack-growth calculations of the article use dN* = 0.625 |
| `[WARN] l0/h_min = ... < 2` | refine the mesh along the crack path or increase l0. The check uses the smallest element edge of the whole model (of a uniform sample of elements for parts with more than 60,000 elements) and does not show that the crack path is resolved; the element size along the expected crack path ($h \leq l_0/2$) must be checked separately |
| `[ERROR] ... Unsupported element type(s)` | use CPE4T, CPS4T, CPE3T or C3D8T |
| `[ERROR] BC ... has no non-zero component in any step (is it a support?) ... Candidates in this model: ...` | a support was selected; choose one of the listed candidates under Fatigue Loading → Load / BC |
| `[ERROR] ... several non-zero ... components` | keep one non-zero component and move the others to another BC |
| `[ERROR] ... is a temperature (phase-field) boundary condition and cannot drive the load` | select the mechanical Load/BC |
| `[ERROR] Base material "..." does not exist` / `No section uses material "..."` | select the material of the section, or `(auto) section for all parts` |
| `[ERROR] Part(s) ... already have a section assignment` | the automatic section is only for parts without a section; select the material of that section |
| `[ERROR] Step ... is a StaticStep` (or another type) | a Coupled temp-displacement step is needed: Step Manager → Replace, or Step mode = Create new use Step mode = Use existing with that step, or set Previous step to it |
| `[ERROR] ... is created in step ..., which comes after Previous step ...` | |
| `[ERROR] Load ratio R must be less than 1.0` | correct R |
| `[ERROR] IPC must be a multiple of 4` | for example 8, 16, 32, 64 or 128 |
| `[ERROR] Bundled ... has been` | restore the released Fortran file |

| Message | Meaning and action |
|---|---|
| `modified` | |

The Check tab lists the checks; the messages appear in the message area at the bottom of the Abaqus/CAE window.

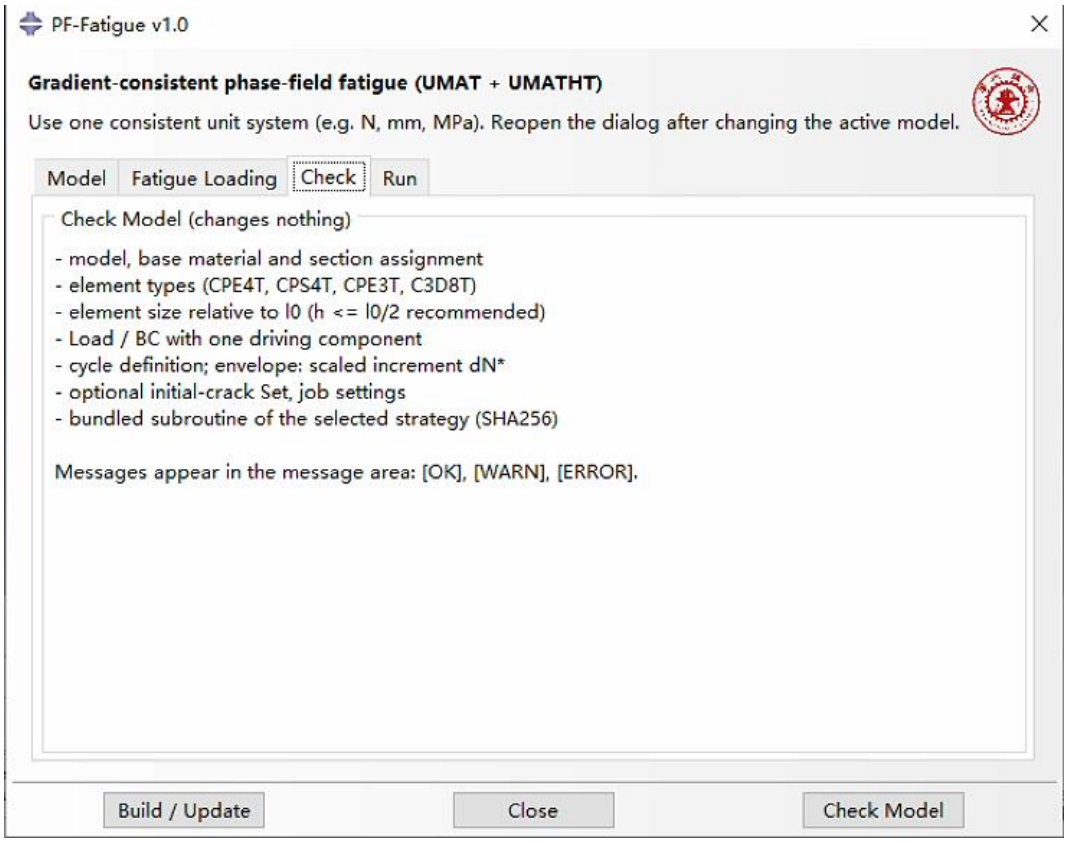



**Fig. 4.** Check tab.

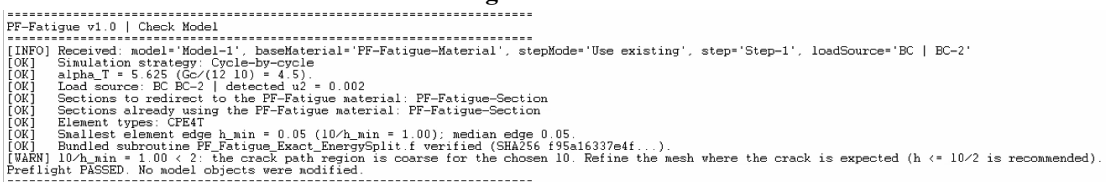



**Fig. 5.** Message area after Check Model. Each check is reported as [OK], [WARN] or [ERROR]. Here the mesh check warns that $l_0$/h_min = 1 < 2, so the mesh is too coarse for the chosen $l_0$.

### 5.4 Run tab

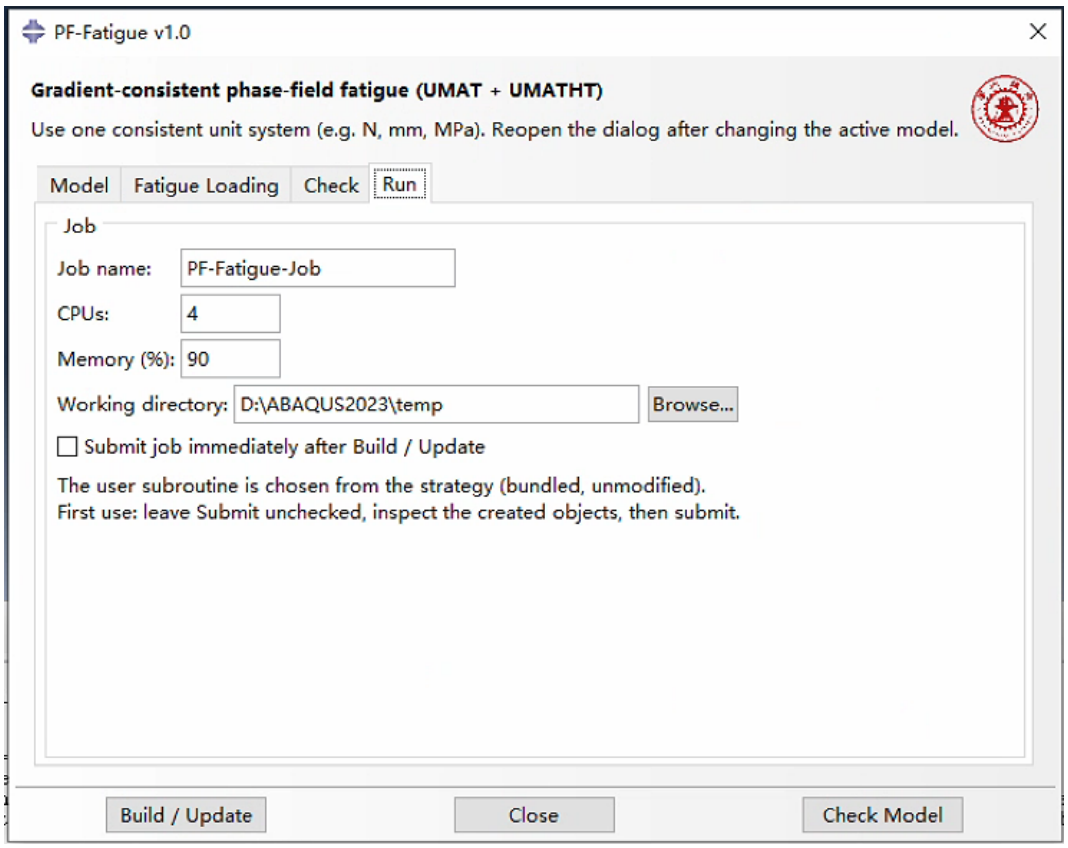



**Fig. 6.** Run tab.

The Run tab holds the job name, the number of CPUs, the memory, the working directory and the option Submit job immediately after Build / Update. For a first model, leave Submit unticked, click Build / Update, inspect the created objects in the model tree, and submit from the Job Manager.

## 6. Strategy and numerical settings

### 6.1 Cycle-by-cycle

For problems with tens to hundreds of cycles. The results depend on IPC, so models are compared at the same IPC. $\alpha_T = Gc/(12l_0)$ is the reference value of the article.

### 6.2 Envelope (dN)

For long lives. $p, \eta_T$ and $\kappa$ are determined from *S-N* or Paris data with the scripts in `postprocessing/identification`. The resolution is set by the scaled increment

$$\Delta N^* = \frac{\Delta N \, \beta^{2\kappa p}}{\eta_T} \tag{1}$$

with $\beta = (1 - R)/2$.

Runs with the same $\Delta N^*$ have the same resolution. A fixed $\Delta N$ is coarser at small $R$ than at large $R$. Check Model reports $\Delta N^*$ and warns above 1. The crack-growth calculations of the article use $\Delta N^* = 0.625$; halving it changed the cycles to a given crack length by less than 3.5%. Halving $\Delta N$ once checks the setting for a new problem.

## 7. Worked example

A 1 × 1 mm plate (CPE4T, seed 0.05 mm) has an initial-crack Set along $y = 0.5$ from $x = 0$ to 0.5. The bottom edge is fixed, and the top edge is displaced vertically by `BC-Top` in a coupled temperature–displacement step `Step-1`.

1. Create the model; `BC-Top`: u2 = 0.001 in `Step-1`. No material or section is needed.
2. Open PF-Fatigue. The dialog proposes Base material = `(auto) section for all parts`, Load / BC = `BC | BC-Top` and Step mode = Use existing, `Step-1`. Model tab: E = 210000; ν = 0.3; $G_c = 2.7$; $l_0 = 0.1$; Energy split = Miehe.
3. Cycle-by-cycle: Strategy = Cycle-by-cycle; alpha_T = 2.25; k_res = 1e-7. Loading tab: Use existing, `Step-1`, `BC | BC-Top`, maximum 0.001, R = 0.1, cycles 2, IPC 8, initial crack ticked.
4. Envelope: Strategy = Envelope (dN); eta_T = 1, p = 1.2, kappa = 1; cycles 20, dN = 1.
5. Check Model → Build / Update → submit.

The generated input contains `*User Material, type=MECHANICAL, constants=7` (cycle-by-cycle) or `constants=9` (envelope), `*Depvar 4`, the temperature BC $\varphi = 1$ with a constant amplitude on the crack Set, and the coupled step with fixed increments (data line `dt, period, dt, dt`). Both variants run in Abaqus 2023 with the requested number of equal increments, and $\varphi = 1$ on the crack Set from the first increment. The mesh is coarse on purpose ($l_0/h = 2$); the model shows the workflow, not a converged result.

Figs. 7–9 show the same workflow for a plate with a seam crack (cycle-by-cycle, $l_0 = 0.05$ mm, IPC = 128, u2 = ±0.002 mm, R = −1). After Build / Update the message area lists the written configuration (Fig. 7), including the PROPS line, the fixed increment and the number of increments. In this run alpha_T was set to 5.625, and the message shows the reference value $G_c/(12l_0) = 4.5$ next to it.

```
======================================================================
PF-Fatigue v1.0 | Build / Update
======================================================================
[INFO] Received: model='Model-1-Seam', baseMaterial='PF-Fatigue-Material', stepMode='Use existing', step='Step-1', loadSource='BC | BC-2'
[OK]   Simulation strategy: Cycle-by-cycle
[OK]   alpha_T = 5.625 (Gc/(12 l0) = 4.5).
[OK]   Load source: BC BC-2 | detected u2 = 0.002
[OK]   Sections to redirect to the PF-Fatigue material: PF-Fatigue-Section
[OK]   Sections already using the PF-Fatigue material: PF-Fatigue-Section
[OK]   Element types: CPE3T, CPE4T
[OK]   Smallest element edge h_min = 0.006733 (l0/h_min = 7.43); median edge 0.01.
[OK]   Bundled subroutine PF_Fatigue_Exact_EnergySplit.f verified (SHA256 f95a16337e4f...).
PF-Fatigue configuration completed.
  Strategy          : Cycle-by-cycle
  Material          : PF-Fatigue-Material  PROPS = 210000, 0.3, 2.7, 0.05, 5.625, 1e-07, 0
  Fatigue step      : Step-1  (period 350.25, fixed increment 0.0078125, 44832 increments)
  Waveform          : triangular from zero, max 0.002, R = -1, IPC = 128
  Load source       : BC | BC-2 | u2
  Field output      : PF-Fatigue-F-Output every 10 increments
  Initial crack     : Skipped
  Job               : PF-Fatigue-Job  (PF_Fatigue_Exact_EnergySplit.f)
  Working directory : D:\ABAQUS2023\temp
  Submitted         : False
======================================================================
```

**Fig. 7.** Message area after Build / Update: checks and configuration summary.

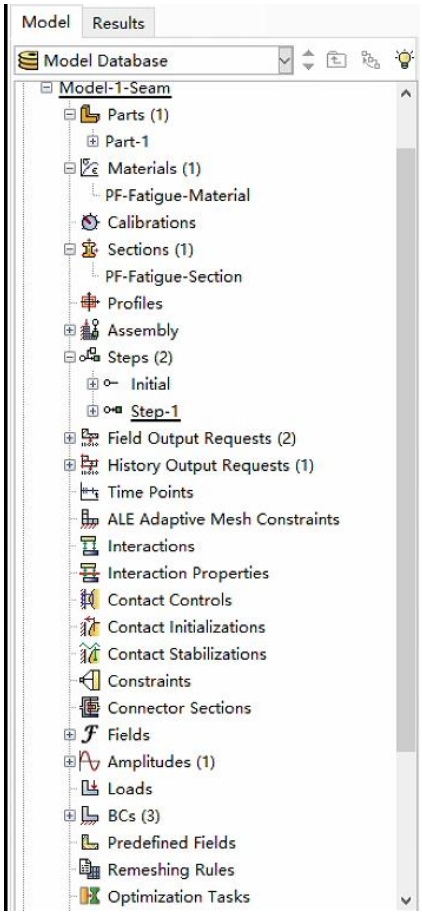



**Fig. 8.** Model tree after Build / Update: PF-Fatigue material and section, coupled step, output requests, amplitude and boundary conditions.

## 8. Results

Table 5 lists the output variables.

**Table 5**: Output variables

| Output | Meaning |
|---|---|
| NT11 | phase field $\varphi$ |
| SDV1 | accumulated fatigue variable ($\bar{\alpha}$ or $\bar{Y}$) |
| SDV2 | history field H |
| SDV3 | previous α (cycle-by-cycle) or Yeq (envelope) |
| SDV4 | degradation function f |

The crack is seen by plotting NT11 in the Visualization module: it is the band where $\varphi \approx 1$. In the cycle-by-cycle strategy the step time is cycles + 0.25, because the first quarter cycle is the ramp to the peak. In the envelope strategy the step time is the cycle number. The status file (`.sta`) must show a constant increment size. The crack length can be extracted with the scripts in `postprocessing/abaqus_python`.

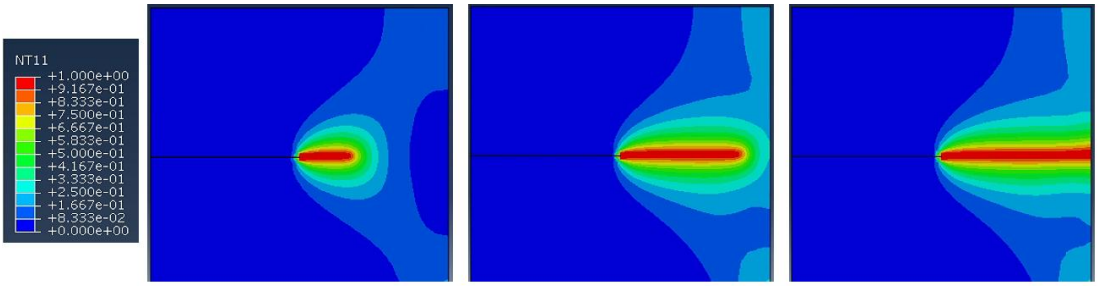



**Fig. 9.** NT11 (phase field $\varphi$) of the seam-crack example at step times 23.4, 36.9 and 60.0 (cycles + 0.25). The crack grows from the tip of the seam to the right edge; the status file shows 7699 equal increments of 1/128 cycle.

## 9. Troubleshooting

**Table 6**: Troubleshooting

| Problem | Solution |
|---|---|
| `LNK1105: cannot close file "standardU.dll"` | the new DLL was briefly locked (simultaneous compilations or a virus scanner) and the linker could not finish; the solver has not started; submit the job again |
| `Problem during compilation` | the Fortran compiler is not linked to Abaqus; run `abaqus verify -user_std` |
| changes to the model are not shown in the dialog lists | close and reopen PF-Fatigue |
| a field is greyed out | it belongs to the other strategy or to an option that is switched off |
| the initial crack is not visible at the start | tick Initial crack in the plug-in and delete any temperature BC or predefined field created by hand (Section 4, item 5) |
| `[WARN] No valid Load/BC was received from the dialog ...; using the only candidate ...` | the dialog did not pass a selection, and the plug-in used the only driving Load/BC of the model; check that it is the intended one |
| the increments in the `.sta` file grow during the step | the step was not built by v1.0; build it again (fixed increments `dt, period, dt, dt`) |
| the crack grows by several elements in one increment | increase IPC, or reduce ΔN (smaller ΔN*) |